

equine Integration

Horse Training Feedback Generator

Project Proposal

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Course:	AI Specialization

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Introduction

Equine Integration supports and advises riders, coaches, trainers, veterinarians, and specialists in equestrian sports, both nationally and internationally. Through innovative measurements during training sessions and competitions, they monitor the physical condition of the horse with precision, developing tailored training programs aimed at optimizing athletic performance in a responsible and health-conscious manner.

A key component of the company's approach is *EI Potential*, a powerful evaluation tool designed by Equine Integration to provide a comprehensive understanding of their client's horse's training, well-being, and performance optimization. Based on scientific research, this app evaluates training sessions and programs, focusing on the unique characteristics of each horse—its discipline, training level, and history. The app offers actionable insights that reveal whether training adjustments are necessary, such as intensifying or easing the workload, adding variety, or maintaining consistency to improve both performance and welfare.

Context

Why?

The purpose of this project is to make the communication between Equine Integration and their customers more efficient, by optimizing the process of the personal generated feedback creation.

What?

In this challenge, the group will optimize the communication between Equine Integration and their customers with an AI model. The company has contact with customers through emails and provides them insights regarding their horse(s) performance. The emails will feature data from EI Potential evaluation tool and will be written in a user-friendly and easy to read format for the customers.

Who?

Name	Occupation	Role	Function
Carolien Munsters	Owner, trainer	Stakeholder	Responsible for providing feedback to the owners of the horses
Zain Farhah	Medio data engineer	Stakeholder	She provided us the dataset and is the contact person for the group in case clarification is needed
Nick Welman	Fontys Teacher	Semestral Coach	Watch over the progress and documentation of the project.
Rink Lycklama à Nijeholt	Fontys Teacher	Semestral Coach	Watch over the progress and documentation of the project.

When?

The project will follow an iterative approach, consisting of three main iterations, with each phase building upon the progress of the previous one. Weekly group sessions will be held every Tuesday to reflect on progress, troubleshoot challenges, and plan next steps.

Project Effort & Timeline

The project will be developed across three key iterations:

- **Iteration 1 (Until 11.05.2025)**
  - Initiate project work.
  - Assign team roles and responsibilities.
  - Define the core problem and research direction.
  - Begin initial data exploration or conceptual development.
  - Submit the first deliverable.

- **Iteration 2 (12.05.2025 – 01.06.2025)**
  - Analyze feedback from Iteration 1.
  - Refine methodology or project scope.
  - Continue development and testing.
  - Submit the second deliverable.

- **Final Iteration (02.06.2025 – 15.06. 2025)**
  - Finalize all components of the project.
  - Prepare presentation materials and documentation.
  - Submit the final iteration and deliver a polished end product.

How?

The end product consists of a web application where our customers will have access to predefined editable feedback message for each of the horse they manage, and an ACWR predicted value.

Domain Understanding

1. Exploratory Research & Initial Understanding

As our stakeholder allowed us considerable freedom in defining the project scope and priorities, much of our domain understanding was self-directed. We were not given specific guidance on what data or variables were most relevant. Instead, we explored the resources provided independently, including access to detailed datasets and a scientific paper explaining *Acute:Chronic Workload Ratio* (ACWR)—a key physiological metric used in sports science to monitor readiness and injury risk.

In addition to the datasets, we consulted academic literature and internal documentation to better understand how workload metrics impact horse performance and welfare. Through this, we learned how monitoring metrics like ACWR helps trainers prevent overtraining and

identify optimal training loads. This insight became foundational to our understanding of the domain.

2. Research Question

Our primary research question evolved around the project's goal:

„How can automated, personalized training feedback improves the decision-making process for horse trainers and enhance the overall value of Equine Integration's services?"

Supporting sub questions include:

- *What types of feedback are most actionable and trustworthy from a business/user perspective?*
- *How can automating feedback improve trainer efficiency and horse well-being, and what impact could this have on user satisfaction and business value?*

3. Methods of Research

Given the absence of hands-on stakeholder interaction, our research methods were primarily based on:

- **Document analysis**, including the ACWR study provided by Equine Integration.
- **Dataset exploration**, examining structured data tables (Horses, Horse_Readiness, eI_User_Annotation, and eI_Communication) to understand how performance and feedback data are recorded.
- **Internal team discussions and exploratory analysis**, where we formulated hypotheses about ACWR and experimented with how it might drive training feedback.

These methods helped us form a conceptual understanding of the connection between workload patterns and feedback suggestions, guiding the direction of our generative feedback solution.

4. Findings and Data Considerations

Our investigation confirmed that **ACWR** is a scientifically grounded metric for determining how prepared a horse is for training or competition. It compares the acute (short-term) workload to the chronic (long-term) workload, offering insight into whether a horse may be overtrained, undertrained, or in an optimal zone.

Key points we learned about ACWR include:

- **High ACWR (>1.5)** may suggest overtraining or risk of injury.
- **Low ACWR (<0.8)** may indicate insufficient training or loss of fitness.
- **Optimal ACWR (0.8 – 1.5)** lies in a range that balances performance readiness with health preservation.

This understanding helped us define our **target variable**:
The qualitative feedback category, generated based on quantitative ACWR and training performance data.
Our system's goal is to produce **natural-sounding, personalized recommendations** using structured input data such as ACWR, activity type, performance analyses, and historical trends.

To do this, we focus on combining:

- ACWR trends from Horse_Readiness
- Performance analysis from eI_User_Annotation
- Historical communication logs from eI_Communication

This holistic view allows our system to produce feedback that not only reflects the horse's current physical state but also aligns with prior coaching patterns.

Analytic Approach

Nature of the Problem

Our client's problem is rooted in the fact that they currently have to spend a lot of time personalizing weekly feedback for their clients. This creates an issue, as it gives them less time to focus on more important tasks for their business, and they have to constantly write messages so their clients can understand how they and their horses are doing.

How Are We Going to Solve It?

We are going to create a system that predicts how the horses' training levels will look in future periods — either over a month or a quarter. This will give the riders/owners a clearer view of what they should expect. As a result, our client will need to write less feedback, reducing their work. It will also lower client stress, increase trust, and give our client more time to focus on more important tasks.

Target Variable

Our target variable will be the **ACWR (Acute:Chronic Workload Ratio)**. This metric compares how much the horse is training this week to how much it has been training over the past few weeks. It helps the trainer avoid injuries by showing if the horse is doing too much too fast or if the training is too light to lead to progress.

Risks

Our current solution includes a few core risks. The biggest one is that horses may sometimes get sick, leading to actual training values being much lower than predicted. Another risk is that if the predicted value is higher than what's achieved, the trainer might be dissatisfied.

Data Requirements

Data Understanding

- Data Source:
  - Datasets for our project were provided by our stakeholder Equine Integration.
- Data Format:
  - Datasets are stored as CSV files.
- Dataset Description:

Table Name	Description
Horses	Contains horse profile details
Horse_Readiness	Contains the ACWR for each of the horses for a specific time period
el_User_Annotations	Contains all activities of the horses
el_Communication	Contains all the communication done for the horses

Horses

Columns Name	Data Type	Description
ID	Integer	Unique ID for a horse
ClientID	Integer	Unique ID for a user

el_Communication

Columns Name	Data Type	Description
HID	Integer	Unique ID for a horse
AID	json	Activity ID for a specific comment. If it is -1, then this comment was made for a horse and is not on a specific activity
DateTime	String	
Comment	String	The comment that was sent
userAccId	Integer	User account ID of the user that sent the comment. If it is -1 then it is a comment sent by Carolien and not the user
Id	Integer	Unique ID for specific comment

Horse_Readiness

Columns Name	Data Type	Description
HorseID	Integer	Unique ID for a horse
ACWR	json	A dictionary which stores the Acute Chronic Workload Ratios (ACWR) for every day
latestItrimpACWR	float	Most recent Itrimp value calculated

el_User_Annotations

Columns Name	Data Type	Description
el_ActivityID	Integer	Unique ID for an activity
CliendID	Integer	Unique Id for a user

Analyses	json	Analysis of the activity (everything we calculate from the raw data)
usercomment	String	Comment of the user on this speci ic activity

Highlighting Features

Valuable metrics identified for predicting the target variable include:

- Average & Max Heartrate
- ACWR value in isolation (without considering its training type)
- Speed
- User comments
- Duration
- Rest time

Our team has selected these columns for initial visualizations and model training. However, this selection may evolve as we conduct deeper data analysis and gain further insights into feature relevance.

Ethnical Concerns

- The dataset is not publicly available and contains sensitive personal information, which raises user privacy concerns. Therefore, special measures must be taken to ensure data security and compliance with ethical standards.
- The following guidelines will be followed:
    - Do not input the data into generative AI tools (e.g., ChatGPT, Copilot) unless proper anonymization and security controls are in place, as this could breach confidentiality agreements.
    - Do not share the data with individuals or parties who have not signed a Confidentiality Agreement specific to the project.
- The data includes personal information related to horses, which must be protected under applicable data protection and ethical handling protocols. All team members are responsible for upholding data security standards throughout the project lifecycle.